

Helix Thready — Cost & Capacity

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Rev	Date	Author	Change
1	2026-07-22	swarm (deployment)	Initial cost quantification: Hetzner host + object tier + backup secondary + GPU node + bandwidth, capacity envelope for three co-resident stacks, and the subscription+metered billing-offset model

The final research request explicitly defers cost quantification to this pack: “*Cost estimates — Hetzner host + object storage + optional GPU; billing offsets via subscription/metered (to be quantified in the deployment pack)*” (§16.3). This document closes that promise. It sizes the **monthly run cost** of the single-host, three-environment topology, ties each line item to the capacity it buys, and specifies the **subscription + metered** billing-offset model that funds it (Q11, §8.1).

Diagram source: sibling under diagrams/ . Rendered PNG/SVG exported via Docs Chain (§11.4.65).

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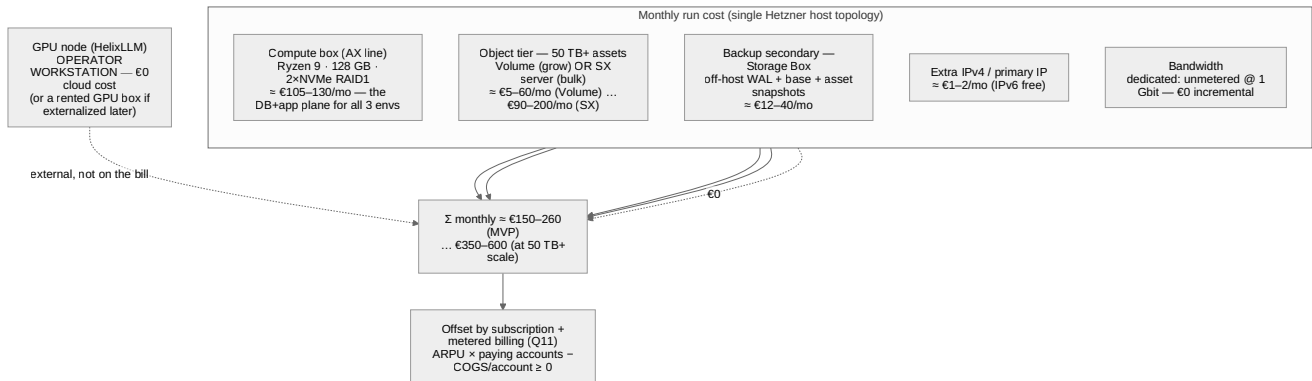
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1. What this costs — the one-line answer

[RESEARCH] [DEFAULT – adjustable] — the **MVP baseline** (one AX-class compute box, assets on a Hetzner Volume, a Storage Box backup secondary, GPU external on the operator workstation) runs at roughly **€150–260 / month**, dominated by the compute box and the asset object tier. The **at-scale** figure (a dedicated SX storage server for a full 50 TB+ asset pool, plus a larger compute box) rises to roughly **€350–600 / month**. All figures are ex-VAT list prices as read during research and **must be re-confirmed at purchase** — Hetzner’s catalogue and prices drift, so the *family + profile* is fixed here, not the cent.

Why a range, not a number. The exact SKU is deliberately [OPEN: host-sizing] (hetzner-provisioning.md §1.1) — the operator confirms the model at purchase after the load test fixes the real pgvector working-set and asset-growth curves. This document gives the envelope those curves land inside.

2. Cost breakdown diagram



Explanation (for readers/models that cannot see the diagram). The monthly bill for the whole Helix Thready deployment is the sum of a small number of Hetzner line items, because the architecture deliberately puts **all three environments on one host** and reaches the expensive GPU as an external endpoint rather than renting it. The dominant item is the **compute box** — an AX-line dedicated server (Ryzen 9, 128 GB RAM, two NVMe drives in RAID1) that carries the Postgres+pgvector data plane and every Go service for dev, sta and prod at once; at roughly €105–130/month it is a fixed cost that does not grow with usage.

The second item is the **object tier** for the 50 TB+ asset store, and it is the one that scales: early on it is a Hetzner **Volume** attached to the compute box (a few euros per terabyte per month, so a modest start is cheap and grows linearly), while at full scale it becomes a dedicated **SX storage server** whose flat price buys tens of terabytes of HDD capacity. The **backup secondary** is a Hetzner **Storage Box** — an off-host SFTP/rclone target that holds the WAL chain, the daily base backups and the asset snapshots for a low flat fee, and whose physical separation from the primary is what makes the RTO target real ([backup-dr.md §6](#)).

Two items are near-free: an extra IPv4 costs a euro or two a month (IPv6 is free), and **bandwidth on a Hetzner dedicated server is unmetered at 1 Gbit**, so serving assets and API traffic adds no incremental cost — a structural advantage over metered clouds for a media-heavy workload. Critically, the **GPU that HelixLLM needs is not on this bill at all**: it runs on the operator workstation / GPU node (Q5) and is reached over the discovery/endpoint layer ([container-topology.md §9](#)), so the cloud cost of the most expensive hardware in the system is zero. The whole sum is then **offset by the subscription + metered billing model** ([§6](#)): the deployment is a cost of goods sold that per-account revenue is designed to cover from day one.

3. Line items

All prices are [\[RESEARCH\]](#) list figures read during research (ex-VAT, EUR) and **must be re-confirmed at purchase**; they are [\[DEFAULT — adjustable\]](#). The point is the *structure* and *relative weight*, not the exact cent.

3.1 Compute host

Candidate	Profile	≈ €/mo	Notes
AX102-class (<i>recommended MVP default</i>)	Ryzen 9, 16C/32T, 128 GB DDR5, 2×~2 TB NVMe	≈ €105–130	Best pgvector/DB latency; assets go to a separate tier
AX52-class (entry)	Ryzen, 64 GB, 2×NVMe	≈ €65–75	Meets the ≥16 vCPU/≥64 GB floor but tight for three co-resident stacks + pgvector
EX-class (Intel)	comparable core/RAM, NVMe	≈ €100–140	Alternative if Intel/ECC is preferred
One-time setup fee	—	€0–39 once	Some SKUs waive it on longer terms

The AX102-class is the **recommended default**; its 128 GB comfortably holds the Postgres shared-buffers + pgvector HNSW working set for prod while dev/sta run throttled ([container-topology.md §7](#)).

3.2 Object tier (50 TB+ assets)

Backing	≈ €/mo	When
Hetzner Volume (attached to the AX box)	≈ €5 / TB / mo (so ≈ €25–60 for 5–12 TB early)	MVP / sub-50 TB start that grows elastically
SX-line storage server (dedicated, on-box HDDs)	≈ €90–200 flat for tens of TB	At/near the full 50 TB+ envelope; cheapest €/TB at scale
Storage Box as live tier	(see 3.3)	Avoid — couples primary+backup; keep it for backups only

Assets are content-addressed and re-downloadable ([backup-dr.md §5](#)), so the object tier can start small and grow with real ingest rather than pre-buying 50 TB.

3.3 Backup secondary

Storage Box tier	Capacity	≈ €/mo
BX11-class	~1 TB	≈ €4
BX21-class	~5 TB	≈ €12
BX31-class	~10 TB	≈ €22
BX41-class	~20 TB	≈ €40

Sized to hold the retention ladder ([backup-dr.md §6](#)): 8 days of hourly WAL + 30 daily/12 weekly/12 monthly base backups + deduped asset snapshots. Because the asset snapshot is content-hash deduped, a 20 TB Storage Box covers a far larger live asset pool.

3.4 GPU node (HelixLLM)

€0 cloud cost in the MVP. HelixLLM runs on the **operator workstation / GPU node** (Q5) and is reached as an **external endpoint** — it is not on the Hetzner bill. If HelixLLM is later externalized to a rented GPU box, add that provider's GPU-hour or monthly rate here as a **separate, elastic** line item; the topology does not change (only the endpoint host does), so this is a pure add-on, not a rework.

3.5 Bandwidth & IPs

- **Bandwidth:** Hetzner **dedicated** servers include **unmetered traffic at 1 Gbit** — the media-heavy asset serving adds **€0** incremental egress, unlike metered hyperscalers. (Hetzner *Cloud* instances meter egress; the dedicated-host choice avoids that.)
- **IPs:** one primary IPv4 is typically included or $\approx$ €1–2/mo; IPv6 is free. The single edge binds the one public IP for all three subdomains ([environments.md §3](#)), so no per-environment IP cost.

4. Capacity envelope (what the money buys)

The one AX102-class box must fit **prod + sta + dev** simultaneously ([container-topology.md §7](#)); this is the capacity the compute line item buys:

Resource	AX102-class has	Prod claims	Sta+Dev ($\approx \frac{1}{3}$ each)	Headroom
vCPU (threads)	32	~ 16 (pg 6 + proc 4 + api/others)	~ 10 combined	~ 6
RAM	128 GB	~ 40 GB (pg 24 + proc 8 + rest)	~ 26 GB combined	~ 62 GB
NVMe (DB + hot)	~ 2 TB usable (RAID1)	pgdata+pgwal hot set	dev/sta small	grows $\rightarrow$ archive old partitions to object tier
Assets	(object tier, §3.2)	50 TB+	shared logical, separated by bucket/prefix	elastic

The headroom is intentional: dev/sta are throttled to $\approx \frac{1}{3}$ so a dev load test cannot starve prod ([environments.md §5](#)). When the prod pgvector working set outgrows RAM, the lever is either a larger box (next AX tier) or archiving cold post-partitions to the object tier ([backup-dr.md §4](#)) — both are cost levers in [§7](#).

5. Three cost scenarios

[DEFAULT — adjustable] — indicative monthly totals for planning:

Scenario	Compute	Object tier	Backup	IP	≈ €/mo total
A — MVP launch (few TB assets, Volume)	AX102 ≈ €120	Volume ~5 TB ≈ €25	BX21 ≈ €12	€2	≈ €160
B — Growth (~15 TB assets)	AX102 ≈ €120	Volume ~15 TB ≈ €75	BX31 ≈ €22	€2	≈ €220
C — At scale (50 TB+ assets, SX tier)	AX102 ≈ €120	SX server ≈ €160	BX41 ≈ €40	€2	≈ €320

GPU is €0 in all three (external). VAT and any one-time setup fees are excluded. Scenario A is the number to budget for the first production cut; C is the ceiling as the asset pool fills the 50 TB+ envelope.

6. Billing-offset model (subscription + metered)

Q11 / §8.1 fix the revenue side as **subscription + metered from day one** — the deployment cost above is the **cost of goods sold (COGS)** that this model must cover. The account/metering machinery itself is the **User Service** ([GAP: #20] , a [BUILD-NEW] container reserved in [container-topology.md §8](#)); this section specifies only the *cost-offset arithmetic*, not the pricing UI.

Offset identity (per month):

```
gross_margin = Σ(account_i revenue) - fixed_infra - Σ(account_i metered_COGS)
where
  account_i revenue      = subscription_tier_price + metered_usage_charges
  fixed_infra             = compute + backup + IP              (≈ €160-320/mo, $5) - amortized
across all accounts
  account_i metered_COGS = object_storage(GB·mo) + LLM_tokens + download_egress(≈€0 on
dedicated) + compute_share
```

- **Fixed vs metered split.** The compute box, backup Storage Box and IP are **fixed** — they do not grow per account, so their per-account cost *falls* as accounts are added. The **object tier** and **LLM token** costs are **metered** — they scale with each account's asset volume and research intensity, so they map cleanly onto per-account metered charges.
- **What to meter** (feeds the metered charge, measured by observability, see [monitoring-observability.md](#)): assets stored (GB·month), LLM tokens consumed (via HelixLLM), posts processed, downloads performed. Egress is *not* metered as a cost because dedicated bandwidth is unmetered (§3.5) — a deliberate margin advantage.

- **Break-even sizing.** With fixed infra $\approx$ €160–320/mo and a modest subscription ARPU, break-even is a small number of paying accounts; every account beyond that improves margin because the largest cost (compute) is already paid. The exact tier prices and metered rates are a product decision owned by the **development** User/Billing service, tracked as `[OPEN: pricing]`.

Anti-bluff note. The metered-COGS inputs are only trustworthy if the metering is real, not stubbed. Usage counters are emitted through the same **observability** pipeline the SLO alerts use, so a broken meter is itself an alertable condition rather than a silent revenue leak.

7. Cost-control levers

Lever	Effect	Reference
Start assets on a Volume , split to SX only near 50 TB	pay for storage as it fills, not up front	\$3.2
Archive cold post-partitions to the object tier	keeps the NVMe hot set (and DB RAM) small → defers a bigger box	backup-dr.md \$4
Dedup asset snapshots	a 20 TB Storage Box backs a much larger live pool	backup-dr.md \$5
Keep the GPU external	the single most expensive component stays off the cloud bill	\$3.4
Unmetered dedicated bandwidth	media egress is free — do not move assets to a metered-egress cloud	\$3.5
Throttle dev/sta to $\approx 1/3$	three environments fit one box → one bill, not three	container-topology.md \$7

8. Verified vs assumed

- **VERIFIED (from the authoritative research docs):** the cost-quantification mandate and its deferral to this pack (`$16.3`); the subscription+metered billing decision (Q11, `$8.1`, `$8.1` billing bullet); the single-host / three-env / GPU-external topology that fixes *what* is on the bill (Q5, Q8); unmetered bandwidth being the reason a dedicated host beats a metered cloud for a media workload (architecture rationale).
- **ASSUMED** / `[RESEARCH]` / `[DEFAULT — adjustable]`: every euro figure (Hetzner list prices drift and must be confirmed at purchase); the AX102-class as the default SKU; the Volume-then-SX object progression; the Storage Box tier sizing; the ARPU/tier prices (owned by the billing service, not this pack).

9. Open items

- `[OPEN: host-sizing]` — exact Hetzner SKU + object-tier backing finalize the compute/object lines after load tests ([hetzner-provisioning.md §1.1](#)).
 - `[OPEN: pricing]` — subscription tier prices + metered rates are a product/billing decision owned by the `development` User/Billing service; this pack quantifies COGS, not the price card.
 - `[OPEN: gpu-externalize]` — if HelixLLM is ever moved off the operator workstation to a rented GPU, add that provider's GPU-hour/monthly rate as a separate elastic line item (topology unchanged).
 - `[OPEN: price-refresh]` — re-read Hetzner list prices at purchase time; the figures here are a research-time snapshot, not a quote.
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